American Computer Science League

2022-2023 ● Contest 2: Binary Counting ● Intermediate Division

PROBLEM: Given a string of characters found on the keyboard, convert each character in the string to the binary equivalent of its ASCII code. In the resulting concatenated string, search for the increasing sequence of binary numbers starting with 0, 1, 10, 11, … until a number cannot be found anywhere in the string. Look from the start of the string. If the binary number is found, remove that occurrence of the binary number from the string. Then look from the end of the string. If the binary number is found, remove that occurrence of the binary number from the string. When the binary number cannot be found at all, output the decimal equivalent of the last binary number that can be found.

EXAMPLE: For the string "Roses are red.", convert it to a concatenated string of binary numbers using each character's ASCII code as follows:

Char	ASCII	Binary
R	82	01010010
o	111	01101111
s	115	01110011
e	101	01100101
s	115	01110011
sp	32	00100000
a	97	01100001

Char	ASCII	Binary
r	114	01110010
e	101	01100101
sp	32	00100000
r	114	01110010
e	101	01100101
d	100	01100100
.	46	00101110

Now search for binary numbers beginning with 0 in the following string:

0101001 00110111 00111001 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00101110

Remove the 0 from both ends so the string becomes:

1010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 0010111

Remove the 1 from both ends so the string becomes:

0<u>10</u>010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00<u>10</u>11

Remove 10 from both ends so the string becomes:

0010 0<u>11</u>01111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00<u>11</u>

Remove 11 from both ends so the string becomes:

0010 0001111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00

Continuing until we search for 1001, the resulting string is:

0000110101011100110000011000010111001001100100000000001000100

The string 1010 can only be found once from the start of the string so the resulting string is:

0000110111001100000110000101110010011001000000000001000100

The process continues until the final string becomes:

0000110000011000010010010000000001000100

The binary number 1101 cannot be found in the final string. The binary numbers 0, 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, and 1100 were found and deleted from one or both sides of the string, but not 1101. The binary number 1100 is the last one found in the string so the output is its decimal equivalent which is 12.

INPUT: A string, *s*, containing any characters found on the keyboard. The string will be fewer than 200 characters.

OUTPUT: For each input, output an integer representing the decimal equivalent of the last binary number that can be found in the string after all deletions have been made.

SAMPLE INPUT:

```
1. Roses are red.
2. A is Alpha; B is Bravo; C is Charlie.
3. A stitch in time saves nine.
4. 1, 2: Buckle my shoe! 3, 4: Shut the door!
5. Is HackerRank the platform used by ACSL?
```

SAMPLE OUTPUT:

```
1. 12
2. 20
3. 14
4. 22
5. 27
```

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TEST DATA

TEST INPUT:

1. The quick brown fox jumped over the lazy dogs.
2. ACSL is 45 years old and going strong.
3. What was the first computer programming language you learned?
4. Supercalifragilisticexpialidocious!
5. Lions and Tigers and Bears, Oh My! This is from The Wizard of Oz.

TEST OUTPUT:

1. 29
2. 27
3. 30
4. 20
5. 26

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问题：给定键盘上的字符串，将字符串中每个字符转换为与其 ASCII 码等值的二进制数。生成的连接字符串中，搜索以 0、1、10、11…开始的递增二进制数数列，直到出现一个在字符串中任意位置都无法找到的二进制数。从字符串的开头进行查找，如果找到二进制数，从字符串中移除这个二进制数。然后从字符串的末尾进行查找，如果找到二进制数，从字符串中移除这个二进制数。直到出现一个再也无法找到二进制数时，输出可以找到的最后一个二进制数的十进制等值数。

示例：对于字符串“Roses are red.”， 运用如下所示每个字符的 ASCII 码，将其转换成二进制数的连接字符串。

字符	ASCII	二进制数
R	82	01010010
o	111	01101111
s	115	01110011
e	101	01100101
s	115	01110011
sp	32	00100000
a	97	01100001

字符	ASCII	二进制数
r	114	01110010
e	101	01100101
sp	32	00100000
r	114	01110010
e	101	01100101
d	100	01100100
.	46	00101110

现在搜索以下字符串中从 0 开始的二进制数：

0<u></u>1010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 0010111<u>0</u>

从字符串的两端删除 0，使字符串变为：

<u>1</u>010010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 0010111

从字符串的两端删除 1，使字符串变为：

0<u>10</u>010 01101111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00<u>10</u>11

从字符串的两端删除 10，使字符串变为：

0010 0<u>11</u>01111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00<u>11</u>

从字符串的两端删除 11，使字符串变为：

0010 0001111 01110011 01100101 01110011 00100000 01100001
01110010 01100101 00100000 01110010 01100101 01100100 00

继续这个过程直到搜索到 1001，生成的字符串是：

0000110101011100110000011000010111001001100100000000001000100

字符串 1010 只能从字符串开头找到一次，因此生成的字符串是：

0000110111001100000110000101110010011001000000000001000100

继续这个过程，直到字符串最终变为：

0000110000011000010010010000000001000100

在最终的字符串中无法找到二进制数1101。二进制数 0、1、10、11、100、101、110、111、1000、1001、1010、1011和 1100 可以在字符串中找到并从字符串的一端或两端删除，但没有找到和删除 1101。二进制数 1100 是字符串中找到的最后一个二进制数，因此输出它的十进制等值数，即12。

输入：包含键盘上任意字符的字符串 S。字符串将少于 200 个字符。

输出：对于每个输入，输出一个整数，表示在所有删除结束之后字符串中可以找到的最后一个二进制数的十进制等值数。

样本输入：

1. Roses are red.
2. A is Alpha; B is Bravo; C is Charlie.
3. A stitch in time saves nine.
4. 1, 2: Buckle my shoe! 3, 4: Shut the door!
5. Is HackerRank the platform used by ACSL?

样本输出：

1. 12
2. 20
3. 14
4. 22
5. 27

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测试数据

测试输入：

1. The quick brown fox jumped over the lazy dogs.
2. ACSL is 45 years old and going strong.
3. What was the first computer programming language you learned?
4. Supercalifragilisticexpialidocious!
5. Lions and Tigers and Bears, Oh My! This is from The Wizard of Oz.

测试输出：

1. 29
2. 27
3. 30
4. 20
5. 26